

Zoo Visitors' Initial Assessment of an Animaloid Robot as a Zoo Exhibit

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Figure 1: Yuki in the zoo

ABSTRACT

Animaloid robots currently have few to no applications other than as a toy in private contexts or as an operational tool in rescue scenarios. A potential further field of use could be zoos, e.g. to offer an additional attraction, to substitute sick animals, to replace already extinct species or to fulfill a certain educational mission. In order to investigate such purposes an animaloid walking robot was exhibited unannounced in an enclosure of a city's large zoo. Interviews with visitors were conducted in order to answer two

research questions: Is an animaloid robot suitable as a zoo exhibit from the visitors' point of view? How do the visitors to a zoo perceive and judge an animaloid robot? The evaluation showed that according to the visitors, the robot, although not perceived as an animal, is considered just as much an attraction and equally suitable as an object of observation in a zoo.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; **Ubiquitous and mobile computing**.

KEYWORDS

animaloid robotics, human-robot interaction, zoo

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1 INTRODUCTION

Due to the constant development of artificial intelligence and the ever-improving advances in robotics, robots are becoming more and more present, even ubiquitous, in our everyday lives.

Animaloid robots, i.e. robots that strongly resemble ordinary animals in their structure and movements, were developed primarily for military purposes [12]. In the meantime, this type of robots can also be purchased on the civilian and private market. However, there seems to be a lack of use cases for animal-like robots. Several institutes are researching areas of application. For example, the German police are examining the use of animaloid robots in dangerous places, such as ruins after a disaster, where they explore the area and possibly find missing civilians [1]. The Robot Museum in Madrid, Spain [13], is not an application, but a historical and typological showcase of robotics. Including but not restricted to animaloid ones, you will find robots of all shapes and sizes there. They are illustrative objects in a gallery-like context that is explicitly and exclusively devoted to robotics. People who visit the museum want to see and expect to see robots.

Our research group had the idea that animal-like robots, in contrast, could well be used as an addition to zoos and be exhibited there. A robot could be an additional attraction or serve as a replacement for an animal that is currently unavailable at the zoo due to illness. An animaloid robot could also, as part of the zoo's educational mission, mimic a specific animal to give visitors the opportunity to experience endangered or even extinct species. Beyond these more purpose-oriented use cases, we wanted to stimulate a reflection on the question of what actually constitutes an animal and to what extent supposedly intrinsic animal appearance and behaviour can be substituted by robots. While there is a long-standing and at the same time current debate with regard to humans and their replaceability by artificial intelligence, the question of what is genuinely animal as opposed to digital is newer and largely unanswered. In this sense a zoo provides an interesting context, since unlike the museum mentioned above, people who visit a zoo want to see and expect to see animals, so that it sparks the desired reflection when they find a robot instead.

The aim of this paper is to answer two research questions. First, is an animaloid robot suitable as a zoo exhibit from the visitors' point of view? Second, how do the visitors to a zoo perceive and judge an animaloid robot? For this purpose, a robot developed by Boston Dynamics called "Spot" [6] was used. Due to its dog-like appearance visually and from its motor skills, this robot offers a good basic prerequisite for the study. In cooperation with the Allwetterzoo in Münster, Germany, the robot was exhibited unannounced in a typical animal enclosure (shown in Figure 1). As with any conventional zoo animal, its enclosure was provided with a species label, and the robot itself was presented all in all like a traditional zoo animal. Several interview questions were used to try to answer the research questions and to gain a first overall impression of the extent to which such robots are suitable in zoos.

2 DESIGN OF USER STUDY

The study was carried out at the "Allwetterzoo Münster" a zoo in North Rhine-Westphalia, Germany. The goal was to present an animaloid robot in the same way as a real animal in an enclosure. The zoo supported the research team by preparing the robot's enclosure like that of a real animal. In addition, the zoo designed a species sign and named the robot dog "Yuki". The exhibition was unannounced, so the visitors initial reaction was captured.

The robot was controlled via remote by a member of the research team who stood in hearing range, but not in the line of sight to create the illusion that spot could be an autonomous robot which reacts to visitors on its own. This limited the differences between Yuki and a real zoo animal to the robot itself. But due to the control by a team member, Yuki was more active towards the visitors. It was less passive as conventional zoo animals. It sometimes reacted to basic orders like "sit" and "stand" of the visitors to create a "magic moment". This leads to a new type of attraction in zoos, Yuki might look like an animal but understands the visitors which results in a new experience.

The survey took place in the form of interviews. For this purpose, a questionnaire was developed, which consisted of four quantitative and four qualitative questions. The quantitative questions are designed as 5-point-likert-items focused on the participants level of agreement towards the suitability of an animaloid robot as a zoo exhibit. At the same time these questions also address the participant's reception of an animaloid robot. The qualitative questions were prepared in free-text format. Here, the visitors were mainly asked about thoughts and feelings addressing their perception of the robot.

The questionnaire with the relations from interview questions to research questions can be viewed in the appendix.

3 RESULTS

A total of 36 people were interviewed ($N = 36$). Eighteen males and seventeen females were interviewed, one person was of different gender. The respondents were assigned to different age ranges. The ranges were children and young people with an age of up to 18 years, adults who are in an age range of 18 to 60 years and seniors who are in an age range of over 60 years. These age ranges correspond with the zoo's various target audiences.

In order to analyze the survey's results two methods were used. The Spearman method was used to find correlations in the quantitative results. It was applied to bivariate variables built from pairing the results of the likert-encoded questions with each other and with additionally collected information about the participants' gender, age and frequency of visits in the zoo. For the evaluation of the qualitative questions, the thematic analysis [5] was used. In total, 54 different interpretive codes could be identified. From these codes, thematic umbrella terms could be determined, with two terms of which are in opposition to each other. Robot and animal—attracting and repulsive.

3.1 Quantitative questions and results:

This subsection shows the results of the quantitative questions. After that, statistical correlations between the quantitative questions

and the respondents' age groups using the Spearman correlation are shown.

When answering the question “I perceive Yuki just like an animal”, there are two main opinions on this question regarding the visitors point of view. On the one hand up to 35 % of the questioned strongly disagree and about 30 % disagree with perceiving Yuki as a real animal. On the other hand, there are more than 20 % that strongly agree to perceive Yuki as a real animal. All in all, we can say that most visitors tend not to see Yuki as a real animal.

In the question “I think Yuki has its justification as a zoo animal, even though biologically Yuki is not an animal”, up to 40 % fully agree. In addition, more than 20 % tend to agree. With that in mind, most people see a justification for animaloid robots especially Yuki in a zoo. On the other hand, a little more than 20 % strongly disagree. The correlation of the relationship between Yuki's right to be a zoo animal and the respondents age group shows, that there is a positive correlation between the two variables, $r(36) = .377$, $p = .024$ (Figure 2 a). The correlation shows significantly higher approval in the two younger age groups than in the oldest age group.

A slightly similar distribution of answers can be seen in answering “I see animaloid robots as a future attraction in zoos”. About 40 % (strongly agree) and 20 % (agree) see animaloid robots as a possible future attraction in zoos. This contrasts with just 20 % who do strongly disagree. We see more people who can imagine this prediction. No significant correlation between Yuki as an attraction in zoos and the respondents' age group could be determined.

When answering “I would have another look at Yuki on my next visit”, most of the visitors can imagine seeing Yuki again at the zoo. More than 80 % would look at Yuki again. None of the visitors would strongly disagree. Only a few people would disagree. This emphasizes a general interest in Yuki and visitors would look for Yuki again at their next zoo visit. The Spearman's rank correlation between this research question and the respondents' age group shows a positive correlation between the two variables, $r(36) = .467$, $p = .004$ (Figure 2 b). Again, the two younger age groups tend to agree significantly more than the older age group.

Statistical correlations could also be found between the quantitative questions themselves. All quantitative questions were compared with the question about the perception of Yuki as an animal. From this (Figure 2), tendencies towards generality are concluded and noted in the form of statements. The Spearman's rank correlation shows a relationship between the perception of Yuki as an animal and Yuki's right to be a zoo animal. There is a positive correlation between the two variables, $r(36) = .450$, $p = .006$ (Figure 2 d). Another Spearman's rank correlation shows a relationship between the perception of Yuki as an animal and Yuki as an attraction in zoos. There is a positive correlation between the two variables, $r(36) = .476$, $p = .003$ (Figure 2 c).

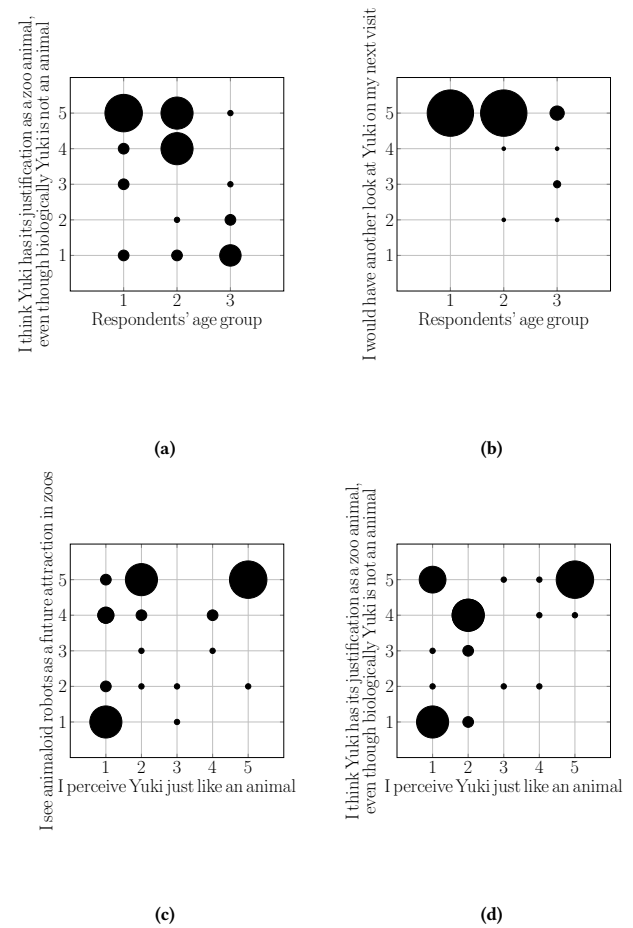


Figure 2: Correlations between different questions and visitor criteria—(a) and (b) with the encoding on x-axis with 1: <18 years; 2: 18–60 years; 3: >60 years (c) and (d) with the encoding on x- and y-axis with 1: strongly disagree to 5: strongly agree

The following statements can be made:

Those who perceive Yuki the same way as an animal will also find that Yuki has a right to be a zoo animal.

Those who don't think Yuki has a right to be a zoo animal don't perceive Yuki the same way as an animal.

Those who perceive Yuki in the same way as an animal will see animaloid robots as an attraction in zoos in the future.

Those who do not see Yuki as an attraction in zoos in the future do not perceive Yuki the same way as an animal.

3.2 Qualitative questions and results:

The first qualitative question given to the respondents was “What were your first thoughts and feelings when you discovered Yuki?”. The initial impressions of Yuki were mostly positive (50 %) to neutral (47 %). Only one respondent had a negative impression of the robot. Some excerpts of these impressions are “Interesting, high quality, [...]”, “Exciting, [...] thrilling, [...] an eye-catcher!”, “Strange, [...]”, “That's the future.”, “[...] does it have to be?”, “Great!”.

The key statements resulting from the question “What makes Yuki interesting for you?” are the technology and progress, as well as the unexpected movements of Yuki. Furthermore, it was a kind of unique attraction for most of the visitors.

A further question was regarding Yuki’s enclosure (as seen in Figure 1): “Do you think the enclosure is suitable for Yuki? Why? Why not? What would you change?”. 15 respondents found the enclosure suitable for Yuki, 11 found it unsuitable and for 7 respondents the answer was unclear. A selection of suggested improvements for the enclosure were more climbing opportunities, more movement related challenges like obstacles, a more futuristic looking enclosure and no water (pond) in the enclosure at all.

In response to the question “Is Yuki a living being to you? Why? Why not? What constitutes a living being for you?”, 82.4 % of respondents answered no, Yuki is not a living being. The reason given was, for example, that Yuki is not biological, but technical and does not possess any independence. In addition, Yuki does not look like an animal, since it lacks a face, fur, sounds and emotions. Furthermore, Yuki would lack a soul, and it does not need to eat food at all. For 14.7 % of respondents Yuki is a living being, because it seems to move independently or autonomously, it has an animal-like shape, and it reacts to the visitors dog-like in form of reacting to specific orders. One respondent gave unclear answers in this regard.

Another question was “What do you think is the biggest difference between Yuki and a classic zoo animal?”. Frequently mentioned differences regarding Yuki’s appearance were that Yuki has no fur and no face. Other additional differences that have been mentioned were the fact that Yuki is not a living being, does not need to eat and is powered by a battery. Therefore, it can no longer function without electricity. In contrast to real animals, Yuki does not move autonomously, but is constantly remote controlled.

Figure 3 shows the relationship between the perception as an animal or robot and an assessment of its suitability as an attraction in a zoo. It can be seen that only two of the respondents tended to describe Yuki as a repulsive robot. All other assessments regarding a suitability as an attraction range from a neutral to a positive picture. Furthermore, it can be seen that the suitability as an attraction also becomes greater with an increasing perception as a robot. Thus, a trend can be seen that the respondents tend to see Yuki mostly as an attraction. The respondents who perceive Yuki as a robot tend to show a greater interest in the sense of an attraction.

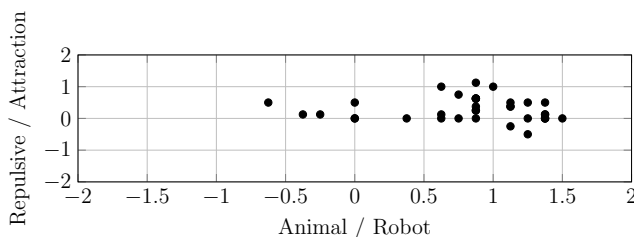


Figure 3: A scatterplot describing the relationship between the perceived animal likeness (x-axes, encoded from animal to robot) and likeability (y-axes, encoded from repulsive to attraction)

4 DISCUSSION

The first question the study aims to answer is whether an animaloid robot is suitable as a zoo exhibit from the visitors point of view. The survey’s results show that the majority of the zoo’s visitors reacted positively to Yuki. Statements like “Interesting!” or “Awesome!” together with general surprise dominated the initial reaction to the robot. However, the results might be influenced by pure curiosity since most of the participants had not seen a robot like Yuki before.

In favor of suitability of a robot as a zoo exhibit speaks that 80 % strongly agree to see Yuki on their next visit and more than 60 % agree or even strongly agree that animaloid robots are eligible as zoo exhibits even though they are no real animals. In addition, the majority of the participants believe that robots populating zoos is a realistic future possibility.

Additional results were found when looking at the answers of those participants aged older than 60 years. Exploring the correlation between age and the answers given from the quantitative questions, it could be seen that with increasing age people are less convinced of a robots’ eligibility as a zoo exhibit and are more likely to skip Yuki’s enclosure on their next visit to the zoo. Older adults resisting technology is a recurring theme addressed several times before in HCI research [7][19][2]. While research in the field of health care robots has shown little difference in the acceptance between different age groups [4][8] this study’s results suggest that for animaloid robots presented in the zoo there is a significant difference.

The second research question addresses the visitors’ perception of Yuki. When asked whether they perceive Yuki just like a real animal more than 60 % disagreed or strongly disagreed. In contrast to the study’s other quantitative questions the perception of Yuki does not correlate with the age. Interestingly, the question was quite polarizing, with few neutral results and most participants either strongly agreeing or disagreeing. Answers to whether the chosen enclosure is suitable for Yuki also support these results. They partly targeted species-appropriate housing implying high animal likeness. However, the majority stated that either the enclosure does not matter since a robot has no needs or suggested changes only increasing the robots’ entertainment value. Naturally these results depend strongly on the chosen robot and cannot be applied to animaloid robots in general. So what does a robot need to be perceived like a real animal? Qualitative results of the study show that—apart from answers stating a robot can never be an animal—the animal likeness can be increased by more natural movement, a greater visual similarity and foremost by autonomic actions.

The thematic analysis of the qualitative results firstly confirms that visitors perceive Yuki more as a robot than a real animal. Secondly it shows that Yuki is seen more as an attraction than as repellent. Looking at the connection between these two scales it can also be said that participants who perceive Yuki as an animal do not consider it repellent.

The resulting assumption that there is a correlation between the perception of the robot as an animal and its suitability as a zoo exhibit is backed by analyzing the correlation of the quantitative results. Perceiving Yuki just like a real animal correlates with its justification as a zoo exhibit as well as assessing robots as future zoo attractions. The more Yuki is perceived as an animal, the higher

the approval of Yuki as a zoo exhibit and the assessment of robots as future zoo attractions.

With this information a connection between the two research questions is shown. The suitability of a robot as a zoo exhibit is highly correlating with whether it is perceived as an animal.

A similar connection has been found in recent research. Examining the connection between animal likeness and likability of zoomorphic robots suggests that there is a quadratic correlation between the two elements creating a minimum of likability for medium animal likeness described as the uncanny valley [10]. The absence of an uncanny valley in this study's results might be caused by the high variance in the participants' age which—as the results show—has a significant impact on their attitude towards Yuki. While the uncanny valley effect focuses on the actual animal likeness this study addresses the individual perception of the robot regarding animal likeness. Perception in turn can not only be influenced by the actual likeness but also by social constructs [15][20][11]. Furthermore, cultural differences are proven to affect robot acceptance [14]. Therefore, the comparatively high acceptance of robots in Germany should be taken into account when making comparisons. Furthermore, when planning the introduction of robots in zoos, a country with a high general robot acceptance should be chosen.

5 CONCLUSION

Motivated by the lack of use cases for animaloid robots in the private and civilian sector, this work explored the possibility of robots populating zoos as exhibits. For this purpose, an animaloid robot was exhibited in a zoo enclosure and a survey was conducted with the aim of determining the perception of animaloid robots and their suitability as zoo animals from the visitors' point of view.

It was shown that it is more likely to be assessed as a suitable zoo exhibit if it is also perceived as an animal.

Because Yuki was perceived largely like a robot, one could assume that it is not suitable as a zoo exhibit. But the study showed the opposite. Despite the little animal-like perception of the robot, the vast majority of visitors rate the robot already as a successful attraction. Only with increasing age the approval of the robot weakened and people did see Yuki slightly less as a zoo exhibit. There is therefore a great potential for robots in zoos, especially if the connection between animal-like perception and suitability from the visitors' point of view is exploited by using very animal-like robots or otherwise influencing the perception of robots.

6 FUTURE WORK

The results of this study show that there is a possibility in the future to exhibit animaloid robots in zoos. To rule out a possible influence of the novelty of this exhibit, i.e. that visitors only find the robot interesting because it is something new, a long-term study would be appropriate. The results of the qualitative questions show that the affection for this type of exhibit can be increased even more, if the robot moves more naturally, visually resembles an animal more closely and, above all, acts fully autonomously. To further develop this theory, more research should be done by experimenting with the appearance and behavior of the exhibited robot. Current results that show the influence of various appearance related aspects of robots on their perception can serve as a basis. Fluid movements

are found to be more popular than mechanical movements [21]. Colors and sounds [9] or even dog tails [3][16][18][17] can be used to convey emotions.

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REFERENCES

- [1] Archyde and Hans-Christian Dirscherl. 2022. German police testing Boston Dynamics Robot dog. <https://www.archyde.com/german-police-testing-boston-dynamics-robot-dog/>
- [2] Belén Barros Pena, Rachel E Clarke, Lars Erik Holmquist, and John Vines. 2021. Circumspect Users: Older Adults as Critical Adopters and Resisters of Technology. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 84, 14 pages. <https://doi.org/10.1145/3411764.3445128>
- [3] Pinhas Ben-Tzvi and Yujiong Liu. 2021. Robots with Tails. *Mechanical Engineering* 143, 6 (11 2021), 32–37. <https://doi.org/10.1115/1.2021-NOV2> arXiv:https://asmedigitalcollection.asme.org/memagazineselect/article-pdf/143/6/32/6808827/me-2021-nov2.pdf
- [4] Mriganka Biswas, Marta Romeo, Angelo Cangelosi, and Ray Jones. 2019. Are older people any different from younger people in the way they want to interact with robots? Scenario based survey. *Journal on Multimodal User Interfaces* 14 (07 2019). <https://doi.org/10.1007/s12193-019-00306-x>
- [5] Victoria Clarke and Virginia Braun. 2016. Thematic analysis. *The Journal of Positive Psychology* 12 (12 2016), 1–2. <https://doi.org/10.1080/17439760.2016.1262613>
- [6] Boston Dynamics. 2023. Spot® - The Agile Mobile Robot. <https://www.bostondynamics.com/products/spot>
- [7] Bran Knowles and Vicki L. Hanson. 2018. The Wisdom of Older Technology (Non)Users. *Commun. ACM* 61, 3 (feb 2018), 72–77. <https://doi.org/10.1145/3179995>
- [8] I. H. Kuo, J. M. Rabindran, E. Broadbent, Y. I. Lee, N. Kerse, R. M. Q. Stafford, and B. A. MacDonald. 2009. Age and gender factors in user acceptance of healthcare robots. In *RO-MAN 2009 - The 18th IEEE International Symposium on Robot and Human Interactive Communication*. 214–219. <https://doi.org/10.1109/ROMAN.2009.5326292>
- [9] Diana Löffler, Nina Schmidt, and Robert Tscharn. 2018. Multimodal Expression of Artificial Emotion in Social Robots Using Color, Motion and Sound. In *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (Chicago, IL, USA) (HRI '18). Association for Computing Machinery, New York, NY, USA, 334–343. <https://doi.org/10.1145/3171221.3171261>
- [10] Diana Löffler, Judith Dörrenbächer, and Marc Hassenzahl. 2020. The Uncanny Valley Effect in Zoomorphic Robots: The U-Shaped Relation Between Animal Likeness and Likeability. 261–270. <https://doi.org/10.1145/3319502.3374788>
- [11] Y. Malhotra and D.F. Galletta. 1999. Extending the technology acceptance model to account for social influence: theoretical bases and empirical validation. In *Proceedings of the 32nd Annual Hawaii International Conference on Systems Sciences*. 1999. HICSS-32. Abstracts and CD-ROM of Full Papers, Vol. Track1. 14 pp.–. <https://doi.org/10.1109/HICSS.1999.772658>
- [12] Ford Moses. 2021. See Spot save lives: fear, humanitarianism, and war in the development of robot quadrupeds. *Digi War* 2 (11 2021), 64–76. <https://doi.org/10.1057/s42984-021-00037-y>
- [13] THE ROBOT MUSEUM. 2023. The Robot Museum. <https://www.therobotmuseum.eu/home/>
- [14] Tatsuya Nomura. 2017. Cultural differences in social acceptance of robots. 534–538. <https://doi.org/10.1109/ROMAN.2017.8172354>
- [15] Harry J. Otway and Detlof Von Winterfeldt. 1982. Beyond Acceptable Risk: On the Social Acceptability of Technologies. *Policy Sciences* 14, 3 (1982), 247–256. <http://www.jstor.org/stable/4531894>
- [16] Ashish Singh and James E. Young. 2012. Animal-Inspired Human-Robot Interaction: A Robotic Tail for Communicating State. In *Proceedings of the Seventh Annual ACM/IEEE International Conference on Human-Robot Interaction* (Boston, Massachusetts, USA) (HRI '12). Association for Computing Machinery, New York, NY, USA, 237–238. <https://doi.org/10.1145/2157689.2157773>
- [17] Ashish Singh and James E. Young. 2013. A Dog Tail for Communicating Robotic States. In *Proceedings of the 8th ACM/IEEE International Conference on Human-Robot Interaction* (Tokyo, Japan) (HRI '13). IEEE Press, 417–418.
- [18] Ashish Singh and James E. Young. 2013. A Dog Tail for Utility Robots: Exploring Affective Properties of Tail Movement. In *Human-Computer Interaction – INTERACT 2013*, Paula Kotzé, Gary Marsden, Gitte Lindgaard, Janet Wesson, and Marco Winckler (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 403–419.

- [19] Ria Vaportzis, Maria Clausen, and Alan Gow. 2017. Older Adults Perceptions of Technology and Barriers to Interacting with Tablet Computers: A Focus Group Study. *Frontiers in Psychology* 8 (10 2017), 1687. <https://doi.org/10.3389/fpsyg.2017.01687>
- [20] Viswanath Venkatesh and Fred Davis. 2000. A Theoretical Extension of the Technology Acceptance Model: Four Longitudinal Field Studies. *Management Science* 46 (02 2000), 186–204. <https://doi.org/10.1287/mnsc.46.2.186.11926>
- [21] Álvaro Castro-González, Henny Admoni, and Brian Scassellati. 2016. Effects of form and motion on judgments of social robots' animacy, likability, trustworthiness and unpleasantness. *International Journal of Human-Computer Studies* 90 (2016), 27–38. <https://doi.org/10.1016/j.ijhcs.2016.02.004>

7 APPENDICES



Figure 4: Yuki's species sign

Table 2: Mapping of qualitative interview questions to research questions.

Interview question	Is an animaloid robot suitable as a zoo exhibit from the visitors' point of view?	How do visitors to a zoo perceive and judge an animaloid robot?
What were your first thoughts and feelings when you discovered Yuki?	-	X
Is Yuki a living being to you? Why? Why not? What constitutes a living being for you?	-	X
What do you think is the biggest difference between Yuki and a classic zoo animal?	-	X
Do you think the enclosure is suitable for Yuki? Why? Why not? What would you change?	-	X

Table 1: Mapping of quantitative interview questions to research questions.

Interview question	Is an animaloid robot suitable as a zoo exhibit from the visitors' point of view?	How do visitors to a zoo perceive and judge an animaloid robot?
I perceive Yuki just like an animal.	-	X
I think Yuki has its justification as a zoo animal, even though biologically Yuki is not an animal.	X	-
I see animaloid robots as an attraction in zoos in the future.	X	-
I would have another look at Yuki on my next visit.	X	-

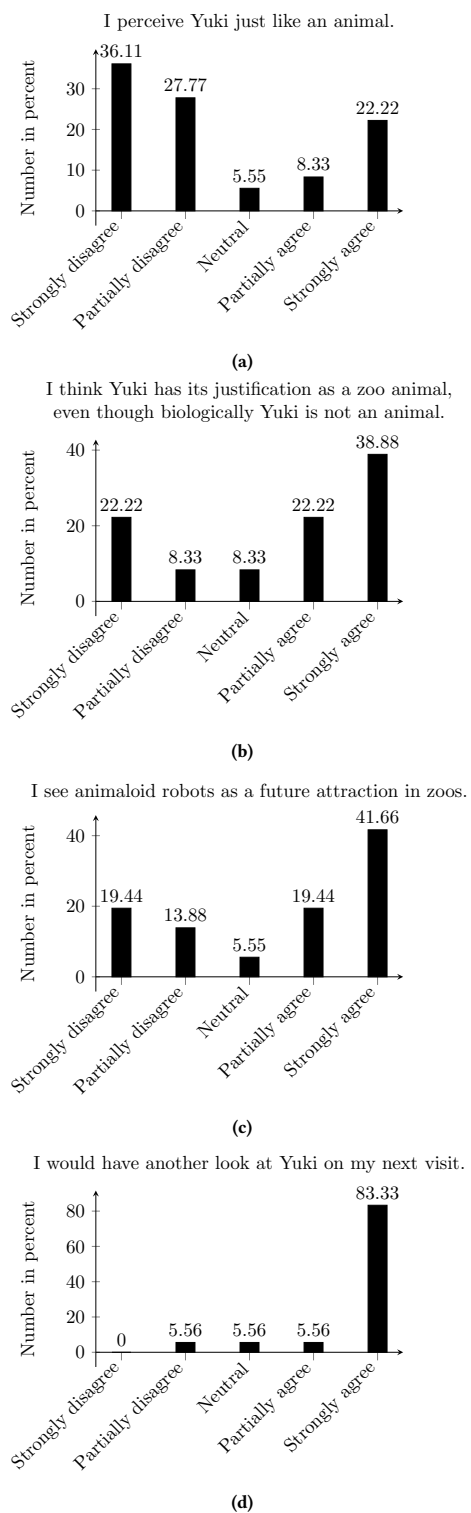


Figure 5: Quantitative results of the survey